Data Flow Diagrams (DFD)

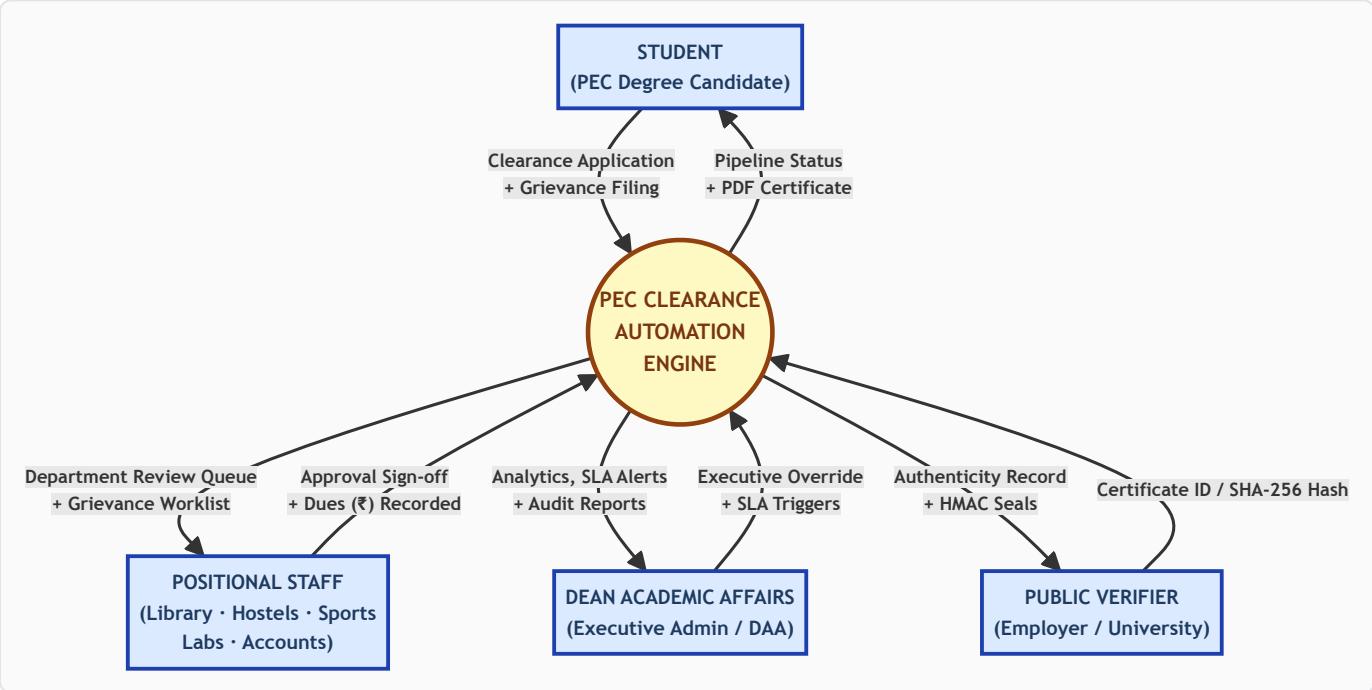
Academic Grievance, No-Dues & Clearance Automation Engine

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Course: Software Engineering
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Notation: Yourdon / De Marco — Circles = Processes, Rectangles = External Entities, Open Boxes = Data Stores

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1. DFD Level 0 — Context Diagram

The Level 0 Context Diagram treats the entire system as a **single process bubble** and maps every interaction between the system and the four external entities on campus.



Level 0 Explanation

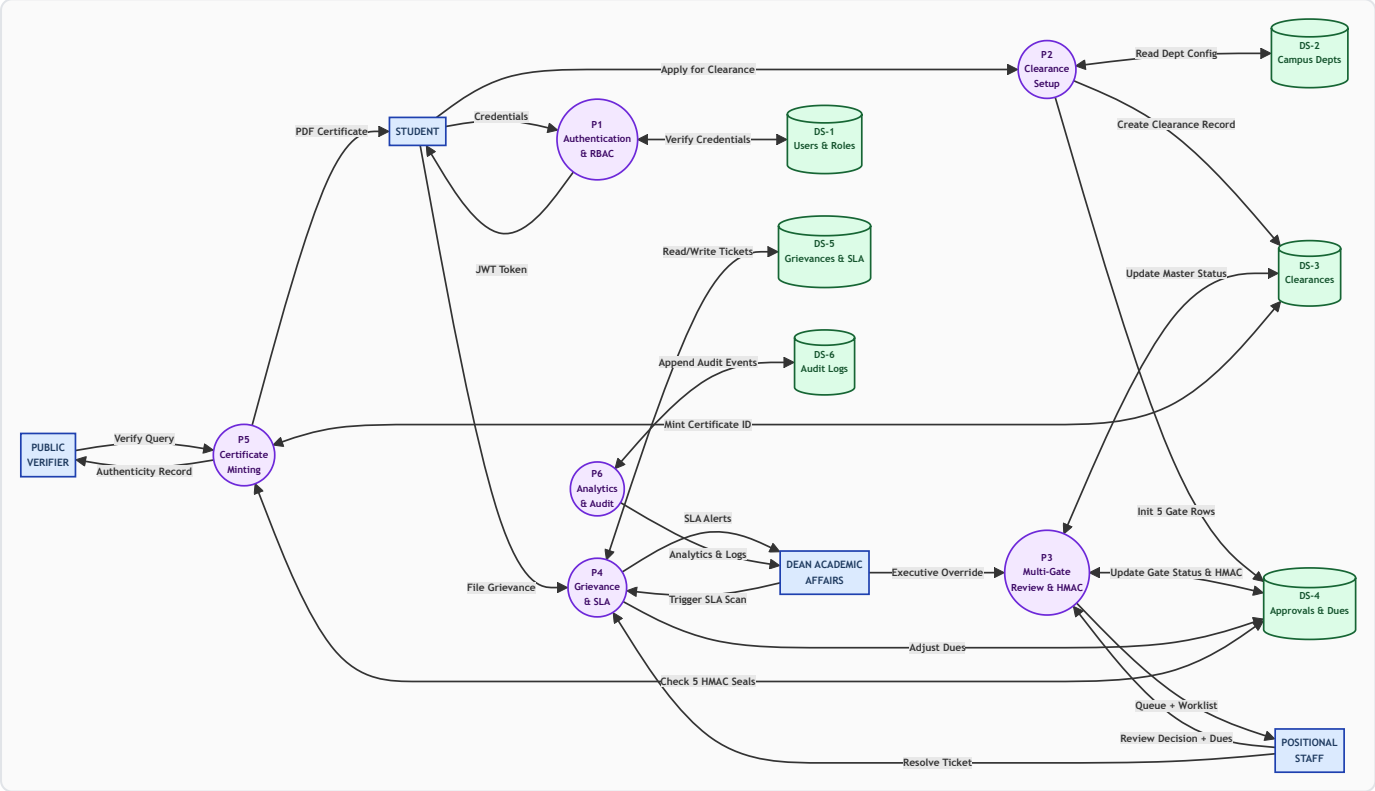
External Entity	Role	What It Sends In	What It Receives Out
Student	Final-year student seeking clearance	Login details, clearance application, grievance complaints	Live clearance status, official PDF certificate
Positional Staff	Officials from Library, Hostels, Sports, Labs, and Accounts	Department approvals/rejections and any pending fine amounts (₹)	List of students waiting for review, grievance tickets to resolve
Dean Academic	Head administrator overlooking campus	Special approval override commands, grievance checks	Overall college clearance reports, alerts for delayed

External Entity	Role	What It Sends In	What It Receives Out
Affairs	clearances		grievances
Public Verifier	Outside company, university, or background checker	Student Certificate ID or QR/Hash code	Instant online confirmation whether certificate is genuine

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2. DFD Level 1 — System Decomposition

The Level 1 DFD decomposes the single engine bubble into **six functional processes** connected by data flows and six persistent data stores.



Process Descriptions

Process	Name	What It Does (Simple Explanation)
P1	User Login & Access Control	Checks student and staff passwords securely; issues a secure digital login token with their role (Student, Staff, or Admin).
P2	Clearance Application Setup	Lets a student submit one active clearance application; creates the application record and sets up the 5 department checklist rows.
P3	Multi-Gate Department Review	Sends applications through Gate 1 (Library, Hostels, Sports), Gate 2 (Labs), and Gate 3 (Accounts) in order; saves digital signatures when approved.
P4	Grievance & Helpdesk Engine	Helps students file complaints if fines or rejections are wrong; sets a resolution deadline and alerts the Dean if delayed.

Process	Name	What It Does (Simple Explanation)
P5	Certificate Generation & Verification	Confirms all 5 departments approved; approves release of ₹10,000 caution money refund; generates the final downloadable PDF certificate.
P6	Reports & Activity Audit Log	Keeps a tamper-proof record of every action (approvals, logins, overrides) and calculates campus clearance statistics for the Dean.

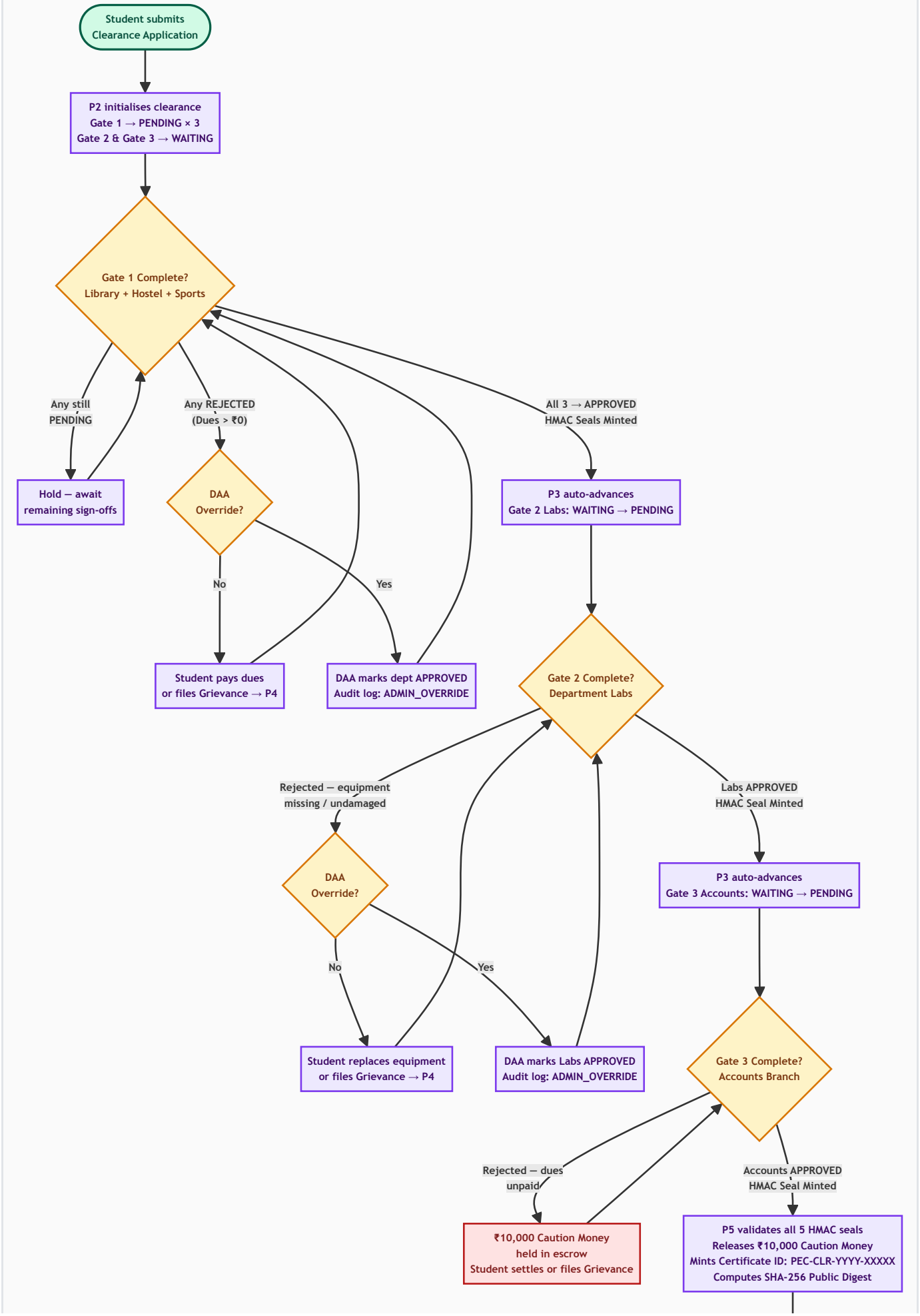
Data Store Descriptions

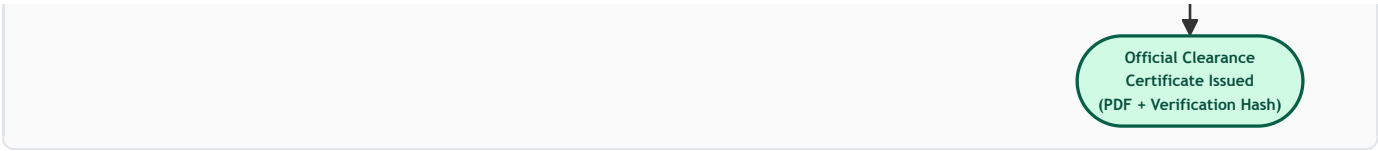
Store	Table	What It Stores (Simple Explanation)
DS-1	`users`	User accounts, encrypted passwords, roles (Student, Staff, Admin), and student roll numbers.
DS-2	`departments`	List of college departments (Library, Hostel, Sports, Labs, Accounts) and their gate order.
DS-3	`clearance_requests`	Master clearance records with overall status (Pending, Approved, Rejected) and certificate details.
DS-4	`department_approvals`	Approval status for each department, unpaid dues/fines in ₹, and digital approval seals.
DS-5	`grievances`	Student complaint tickets, priority levels, resolution deadlines, and staff responses.
DS-6	`audit_logs`	Security history log recording who did what action, to which record, and at what exact time.

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3. Core Decision Tree — Multi-Gate Clearance Pipeline

The Decision Tree traces the **deterministic gate-by-gate path** from a student's initial application to the final Clearance Certificate, capturing every branch condition enforced by the Chain of Responsibility engine.


Student submits Clearance Application
P2 initialises clearance Gate 1 → PENDING × 3 Gate 2 & Gate 3 → WAITING
Gate 1 Complete? Library + Hostel + Sports
Any still PENDING
Hold — await remaining sign-offs
Any REJECTED (Dues > ₹0)
DAA Override?
No
Student pays dues or files Grievance → P4
Yes
DAA marks dept APPROVED Audit log: ADMIN_OVERRIDE
All 3 → APPROVED HMAC Seals Minted
P3 auto-advances Gate 2 Labs: WAITING → PENDING
Gate 2 Complete? Department Labs
Rejected — equipment missing / undamaged
DAA Override?
No
Student replaces equipment or files Grievance → P4
Yes
DAA marks Labs APPROVED Audit log: ADMIN_OVERRIDE
Labs APPROVED HMAC Seal Minted
P3 auto-advances Gate 3 Accounts: WAITING → PENDING
Gate 3 Complete? Accounts Branch
Rejected — dues unpaid
₹10,000 Caution Money held in escrow Student settles or files Grievance
Accounts APPROVED HMAC Seal Minted
P5 validates all 5 HMAC seals Releases ₹10,000 Caution Money Mints Certificate ID: PEC-CLR-YYYY-XXXXX Computes SHA-256 Public Digest



Decision Tree Explanation

The three-gate pipeline enforces **strict sequential dependency barriers**:

- **Gate 1 — Parallel Departmental Stage:** Library, Hostels, and Sports departments operate concurrently. Any rejection blocks further pipeline advancement and routes the student to either pay dues or file a formal grievance (handled by Process 4). The Dean Academic Affairs (DAA) holds an executive override authority at every gate for cases of administrative error or approved waivers, with a mandatory audit trail entry recorded in DS-6.

- **Gate 2 — Academic Labs Stage:** Automatically unlocked only when all three Gate 1 facilities are `APPROVED`. The HOD of Department Labs verifies equipment returns; approval seals the record with an HMAC-SHA256 signature. Any rejection loops back unless overridden by the DAA.

- **Gate 3 — Accounts & Certification Stage:** Automatically unlocked only after Gate 2 is `APPROVED`. The Accounts Branch reconciles all outstanding institutional dues. Upon approval, the ₹10,000 Caution Money deposit is released, Process 5 validates all five HMAC seals, and the system generates the unique Certificate ID and public SHA-256 verification digest — producing the official, tamper-evident PDF clearance certificate.